

Wireless Quick-Dial Cards

Setting an assigned frequency on transmitter & portable receiver

FREE-TUNE TO ANY MHz

Free-tune to any MHz (user bank / Tune / Frequency field).

PICK FINE GROUP FIRST

You must select a **fine / unlocked group** first, then set the freq (Sony, Wisycom).

FREE-TUNE WITHIN BLOCK

Free-tune within a fixed hardware **block**; TX and RX must share the block (Lectro).

CHANNEL CHART ONLY

Channels only — pick from the published **channel chart**; no arbitrary MHz dial (Comtek).

FAMILY (TX / RX)	USE THIS CARD	HOW TO REACH ANY MHz
Sennheiser Evol G3 / G4 (SK/SKM)	Sennheiser evolution G3 / G4	FREE-TUNE TO ANY MHz
Sennheiser EW-DX (SK/SKM-S)	Sennheiser EW-DX	FREE-TUNE TO ANY MHz
Sennheiser EM6000 / SK(M) 6000	Sennheiser Digital 6000	FREE-TUNE TO ANY MHz
Sennheiser 2000 (SK/SKM/SKP 2000)	Sennheiser 2000 Series	FREE-TUNE TO ANY MHz
Sennheiser 3000/5000 (SK 5212 / SKM 5200)	Sennheiser 3000 / 5000 Series	FREE-TUNE TO ANY MHz
Shure ULXD (ULXD1 / ULXD2)	Shure ULX-D	FREE-TUNE TO ANY MHz
Shure SLX-D (SLXD1/2/5)	Shure SLX-D	FREE-TUNE TO ANY MHz
Shure Axient AD / ADX	Shure Axient Digital	FREE-TUNE TO ANY MHz
Shure UHF-R (UR1/UR2)	Shure UHF-R	FREE-TUNE TO ANY MHz
Shure PSM-1000 (P10T / P10R+)	Shure PSM 1000	FREE-TUNE TO ANY MHz
Sony UWP / UWP-D (UTX...)	Sony UWP-D	PICK FINE GROUP FIRST
Sony DWX (DWR / DWT)	Sony DWX Digital	PICK FINE GROUP FIRST
Wisycom MTP / MCR	Wisycom MTP / MCR	PICK FINE GROUP FIRST
Lectrosonics block units (SMV/SMQV...)	Lectrosonics Digital Hybrid	FREE-TUNE WITHIN BLOCK
Lectrosonics wideband (LT / LR “3 Block”)	Lectrosonics L-Series wideband	FREE-TUNE WITHIN BLOCK
Lectrosonics IFB (IFBT4 → IFBlue)	Lectrosonics IFB (IFBlue)	FREE-TUNE WITHIN BLOCK
Lectrosonics Duet (M2T → M2R / M2Ra)	Lectrosonics Duet (M2T / M2Ra)	FREE-TUNE TO ANY MHz
Lectro DBSM/DBSMD (D2)	Lectrosonics D Squared (DBSM / DSQD)	FREE-TUNE WITHIN BLOCK
Comtek 216 (BST-25/216 → PR-216)	Comtek 216 (BST-25 / PR-216)	CHANNEL CHART ONLY
Sound Devices Astral (A20-TX/Mini/HH)	Sound Devices Astral (A20)	FREE-TUNE TO ANY MHz
Zaxcom ZMT (ZaxNet)	Zaxcom (ZMT4.5 / URX100)	FREE-TUNE TO ANY MHz
Crew's WWB rejects assigned freqs	Shure Wireless Workbench (WWB 6 & 7)	SOFTWARE — CALC ONLY
Teradek video (Bolt / Ranger)	Teradek Bolt / Ranger (wireless video)	LOCK TO ONE CHANNEL
5/6 GHz channel ≠ MHz lookup	5 / 6 GHz Channel Chart	REFERENCE CHART
DJI / Rode / Hollyland 2.4 GHz mics	2.4 GHz Auto-Hop Mics	CANNOT BE COORDINATED
Generic / unspecified model	— verify the actual hardware at check-in	VERIFY

Four rules that prevent most check-in failures: (1) **never power up an uncoordinated transmitter with RF on** — put it in its RF-off / standby / mute state first (see each card's red **RF OFF FIRST** note), set the assigned frequency, *then* enable RF; (2) transmitter and receiver must be in the **same band / block**; (3) on Sony UWP-D & Wisycom, select the **fine / unlocked group** before an arbitrary frequency is reachable (Sony DWX also offers direct FREQ INPUT); (4) on digital systems an **IR-sync** (RX→TX) copies the frequency onto the other unit — re-sync after band or encryption changes.

Sennheiser evolution G3 / G4

Analog UHF · portable (camera) receiver shown

FREE-TUNE TO ANY MHz

Step 25 kHz

Range Tune only inside the unit's band letter (A 516–558, G 566–608, Aw+ 470–558...)

Typical report label: Sennheiser EW G2/G3/G4

Sennheiser Evol G3 A

Sennheiser Evol G3 G

Sennheiser Evol G4 Aw+

TRANSMITTER

SK / SKM (bodypack / handheld)

- 1 Press **SET** → **Advanced** → **Tune** → **SET** (navigate with **UP**/**DOWN** + **SET**). Unit switches to channel 1 of user bank **U1** (U1–U6); TX RF is deactivated while in Tune.
- 2 **UP**/**DOWN** to dial the assigned **MHz** (25 kHz steps; hold to scroll). **SET** to store.
- 3 **ESC** back to home; confirm the MHz on screen.

PORTABLE RECEIVER

EK 100 G3 / G4 (camera receiver)

- 1 Press **SET** → **Advanced** → **Tune** → **SET** (switches to channel 1 of bank **U1**).
- 2 **UP**/**DOWN** to the same **MHz**; **SET** to store.
- 3 Or set the RX here, then **SYNC** the transmitter to it via IR.

RF OFF FIRST

Entering **Tune** deactivates the TX RF while you dial ("the RF signal is deactivated", per manual) — tune before approaching the rig. Where fitted, set the mute switch function to **RF Mute** (menu: Mute Mode) for a hard mute.

RF POWER

G4 300/500: **10 / 30 / 50 mW** (Low/Standard/High), **Advanced** ► **RF Power**. G3 300/500: **10 / 30 mW** (Low/Standard) only. 100-series: fixed, max 30 mW.

MATCHING TX & RX

Bank presets (1–20) only hold fixed compatible channels — use **Tune** (bank U) to reach an arbitrary freq.

WATCH OUT

- RX and TX must be the same band letter, or they can't meet on frequency.
- G3 and G4 tune identically. Reports often lump **G2** in too — same family drill, but verify menu names on older G2 units at the table.

SOURCES [ew G4 system manual](#) · [SK 500 G3 manual](#) · [EK 100 G3 manual](#)

Sennheiser EW-DX

Digital UHF · no portable RX in-series — EW-DP EK is the camera option

FREE-TUNE TO ANY MHz

Step fine (MHz + kHz entry; factory presets sit on a 600 kHz grid)

Range US: Q1-9 470.2–550 · R1-9 520–607.8 · V5-7 941.7–959.65 (licensed STL)

Typical report label: **Sennheiser EW-DX Q1-9** **Sennheiser EW-DX R1-9** **Sennheiser EW-D Q1-6**

TRANSMITTER

SK / SKM(-S) (bodypack / handheld)

- 1 Press **SET** to open the menu → **Frequency**;
UP/**DOWN** sets the MHz range → **SET** →
UP/**DOWN** sets the kHz → **SET** to save.
- 2 Or sync from the receiver: press **CH 1**/**CH 2**
on the EM 2 → **SYNC** on the receiver →
short-press **SYNC** on the TX (**Bluetooth LE**
— **no IR**).
- 3 Keep that TX press **short** — holding **SYNC** >
2 s enters firmware-update mode and cancels
the sync.

PORTABLE RECEIVER

EW-DP EK (camera receiver, cross-compatible)

- 1 **UP**/**DOWN**/**SET**/**ESC** → **FREQUENCY** (or
AUTO SCAN / CHANNEL).
- 2 Set the assigned **MHz**; **SET** to save.
- 3 Must be the same **frequency-range variant**
as the TX; EW-DX extras (encryption, LD)
drop off on EW-DP.

RF OFF FIRST

No RF-mute boot combo. **SK / SKM-S**: set the mute-switch function to **RF Mute** (TX menu: **Mute Button**; or RX: Ch → Mute Mode) and slide to mute *before* power-on. Plain SKM has no mute switch.

RF POWER

Fixed **10 mW ERP** (12 mW in Y1-3) — there is **no RF-power menu**.

MATCHING TX & RX

Sync runs over **Bluetooth LE** from the receiver — receiver values overwrite the TX. Manual TX entry and the Smart Assist app also work.

WATCH OUT

- Enabling **AES-256** (System → Link Encryption) requires re-syncing every TX.
- **Link Density** mode needs a receiver restart; presets shift to a 300 kHz grid.
- TX and RX must share the frequency-range variant (Q1-9 vs R1-9...).
- **EW-D** (non-DX) TXs have **no display at all** — set the freq on the EW-D EM (or Smart Assist app) and push with the **SYNC** buttons (BLE). Fixed 10 mW. Q1-6 = 470.2–526; “S1-7” = 606.2–662 (mostly not US-legal).

SOURCES [EW-D/DX/DP combined manual](#) · [EW-DX online manual](#) · [Frequency ranges](#) · [EW-D manual](#)

Sennheiser Digital 6000

Digital UHF · portable (camera) receiver shown

FREE-TUNE TO ANY MHz

Step 25 kHz

Range Up to 88–184 MHz tuning, depending on unit

Typical report label: **Sennheiser EM6000 A1-4****TRANSMITTER****SK 6000 / SKM 6000**

- 1 On the TX, press **SET** to open the menu → **Tune**.
- 2 Set the assigned **MHz** (25 kHz steps); confirm — saving auto-assigns it to preset **U**.
- 3 Match the **transmission mode** (e.g. LR) to the receiver, or the link won't form.

PORTABLE RECEIVER**EK 6042 (2-ch camera receiver, LR)**

- 1 Press the channel button → **CH Settings** → **Frequency** → **Tune**.
- 2 Set the same **MHz**; confirm.
- 3 Then **Sync** transfers the frequency to the transmitter over IR.

RF OFF FIRST

SK 6212: hold **ON/OFF** at power-up until the LED lights red → boots in **RF MUTE**. SK 6000 / SKM 6000 have no RF-mute boot — set the freq on the EM 6000 / EK 6042 and **IR-sync before the pack goes live**.

RF POWER

Set by **mode**: LR — SK/SKM 6000 **25 mW**, SK 6212 **15 mW** (Low 3.5). LD — 3.5 mW (SKM 6000: 1 mW). Mode is set on the **EM 6000** and pushed by sync — not a free dial.

MATCHING TX & RX

On the **EM 6000**, **Sync Settings ▶ Frequency Only** pushes just the freq. The EK 6042 sync always transfers the frequency (long-press the sync button for IR).

WATCH OUT

- Digital 6000 TX work with EK 6042 in **Long-Range (LR)** mode — set both to LR.
- EM 6000 is the rack receiver; EK 6042 is the portable equivalent.

SOURCES [Digital 6000 manual](#) · [EK 6042 manual](#)

Sennheiser 2000 Series

Analog UHF (high-band) · portable (camera) receiver shown

FREE-TUNE TO ANY MHz**Step 25 kHz**

Range Up to 75 MHz switching bandwidth; tune inside the unit's band (e.g. Aw, Gw, Bw). 25 kHz steps.

Typical report label: **Sennheiser EM 2000** **SK 2000 / SKM 2000**

TRANSMITTER

SK 2000 / SKM 2000 / SKP 2000 (bodypack / handheld / plug-on)

- 1 Press **SET** → **Advanced** → **Tune** (SK 2000 = SET + UP/DOWN buttons; only the SKM 2000 has a jog-style multi-function switch). Pick a **user bank (U1-U6)**; fixed banks 1-20 are intermod presets only.
- 2 Set the channel and dial the assigned **MHz** (25 kHz steps); press **SET** to store.
- 3 Easier: set it on the receiver and IR-sync down to the TX (right).

PORTABLE RECEIVER

EK 2000 (camera receiver)

- 1 Press **SET** → **Advanced** → **Tune**; select a **user bank (U1-U6)** and channel.
- 2 Set the same assigned **MHz** (25 kHz); store the setting.
- 3 Hold the RX IR window to the TX IR window → **Sync** (main menu): the freq transfers to the TX automatically.

RF OFF FIRST

RF Mute: hold **ON/OFF** at switch-on until **"RF Mute Off?"** shows → **UP**/**DOWN** to **"RF Mute On?"** → **SET**. Tune/sync, then un-mute the same way.

RF POWER

10 / 30 / 50 mW (US: up to 100 mW) — **Advanced ▶ RF Power** on the TX.

MATCHING TX & RX

EM 2000 / EK 2000 → **Advanced ▶ Sync** transfers the frequency to the transmitter over IR. Or set both by hand to the same bank / channel / MHz.

WATCH OUT

- RX and TX must share the same band / range, or they can't meet on frequency.
- Fixed banks 1-20 are presets only — use a **user bank (U1-U6)** to land on an arbitrary coordinated MHz.
- 2000 series sits above ew G3/G4 and below 3000/5000 — same 25 kHz raster.

SOURCES [SK 2000 manual](#) · [EK 2000 manual](#)

Sennheiser 3000 / 5000 Series

SK 5212-II bodypack · SKM 5200-II handheld · SKP 3000 plug-on — EK 3241 portable receiver

FREE-TUNE TO ANY MHz**Step** 5 kHz — finest on these cards

Range Tune within the unit's switching bandwidth (EK 3241 ≈ 36 MHz; SK 5212-II up to 184 MHz); stay in the legal range.

Typical report label: **Sennheiser EM 3732 / 3732-II** **SK 5212(-II)** **SKM 5200(-II)**

TRANSMITTER

SK 5212(-II) / SKM 5200(-II) / SKP 3000
(bodypack / handheld / plug-on)

- 1 Deactivate the autolock if on. Open the menu → **CHAN**: pick a **VAR-bank** channel (**FIX**-bank presets cannot be changed).
- 2 Go to **TUNE** and set the assigned **MHz** — tunes in **5 kHz** steps; press **SET** to store ("STORED").
- 3 Set output in **POWER**, then EXIT. (TUNE / CHAN only take effect once you **SET** them.)

PORTABLE RECEIVER

EK 3241 (camera receiver)

- 1 Use **CHAN** to pick a channel in bank "**U**" (variable bank, 20 channels U.01-U.20; bank "F" is 32 factory presets).
- 2 Open **tunE** and set the same **MHz** by hand (5 kHz steps); press **SET** ("StorEd").
- 3 **No IR sync on this receiver** — confirm the MHz matches the TX exactly. (Use an EM 3732 rack if you need IR sync.)

RF OFF FIRST

No standby-off boot — set the freq in **TUNE** (and a low **POWER** level) before the pack is near other rigs; confirm, then raise power / go live.

RF POWER

SK 5212-II **POWER** menu: **50 mW** (PWR.HI) / **10 mW** (PWR.LO) / 10 mW low-intermod (**LoI**).

MATCHING TX & RX

The EK 3241 camera receiver has **no IR transmitter-sync** — match the frequency by hand on both ends (5 kHz). An EM 3732 rack can IR-sync the transmitters if you have one.

WATCH OUT

- FIX-bank presets cannot be changed; use the **VAR bank** to land on a coordinated MHz.
- EK 3241 has no transmitter IR-sync — match by hand (or sync from an EM 3732 rack).
- TX and RX must share the same range / band.

SOURCES [SK 5212-II manual](#) · [EK 3241 manual \(US\)](#) · [EM 3732-II manual](#)

Shure SLX-D

Digital UHF · SLXD5 is the portable receiver

FREE-TUNE TO ANY MHz

Step 25 kHz (JB band 125 kHz)

Range US: G58 470–514 · H55 514–558 · J52 558–602 + 614–616 · wideband “+” models G57 470–608

Typical report label: **Shure SLXD4 G58** **Shure SLXD4Q+ G57** **Shure SLXD5**

TRANSMITTER

SLXD1 / SLXD2 (bodypack / handheld)

- 1 Transmitters have **no frequency menu** (menu = RF power, offset, locks) — the frequency comes from the receiver.
- 2 Tune the receiver (right), align the **IR ports**, press the receiver's **sync** button → the TX takes the freq.
- 3 **No RF-mute boot exists** on SLX-D — power packs up away from the rig, sync, then move in.

PORTABLE RECEIVER

SLXD5 (portable slot/bag receiver)

- 1 Press **Enter** from home → **Frequency** → **Manual Setup** (or Channel / Group Scan).
- 2 **▲▼** arrows to set G / C or the exact **MHz** (25 kHz); **Enter** to apply.
- 3 Align IR ports with the TX and press **Sync**.

RF OFF FIRST

Nothing to hold at boot — this family has **no RF-mute power-on** (unlike ULX-D). Set the receiver first, sync at a distance, then approach.

RF POWER

TX menu **RF Power: 1 / 10 mW** (SLXD3 plug-on: 30 mW); or push from the SLXD5 (**TX ▶ RF Power**, Low 1 / High 10) via IR sync.

MATCHING TX & RX

SLXD4 path: push the **control knob** → **Frequency Setup ▶ Manual Frequency Setup** → set G / C / MHz → **Apply**. “+” receivers pair with legacy TXs when **RF Band** matches (Device Configuration ▶ RF ▶ RF Band).

WATCH OUT

- Legacy SLX-D has **no encryption**; SLX-D+ adds AES-256.
- G65 and L57 are **not US bands** — US stock is G58 / H55 / J52 (+ G57 wideband).
- Keep all receivers in one band on the same group.

SOURCES [SLX-D user guide](#) · [SLXD4Q+ user guide](#)

Shure ULX-D

Digital UHF · (no portable receiver — half-rack ULXD4 shown)

FREE-TUNE TO ANY MHz**Step** 25 kHz**Range** Band on the label must match (e.g. G50 = 470–534 MHz)**Typical report label:** **Shure ULXD4 G50****TRANSMITTER****ULXD1 / ULXD2 (bodypack / handheld)**

- 1 ULXD1/2 have **no frequency menu** — the TX menu covers mute mode, mic offset and locks only. The frequency comes from the receiver.
- 2 Tune the receiver channel (right), align IR windows, press **SYNC** → the TX takes the frequency.
- 3 Need RF silence meanwhile? Hold **exit** while powering the TX on until **RF MUTED** shows.

PORTABLE RECEIVER**ULXD4 (half-rack receiver)**

- 1 Open the channel's **RADIO** screen; push the **control wheel** to edit.
- 2 Rotate to the **Frequency** field and set the assigned **MHz** directly; **ENTER** to save.
- 3 Align IR windows, press **SYNC** → **Sync Success** copies it to the TX.

RF OFF FIRST

Hold **exit** during TX power-on until **RF MUTED** displays. Sync the assigned freq from the receiver, then **restart the TX** to un-mute.

RF POWER

1 / 10 / 20 mW — pushed from the receiver as an IR-sync preset (Lo 1 / Nm 10 / Hi 20), not set on the TX.

MATCHING TX & RX

In the **RADIO** menu you can set Group/Channel *or* edit the **Frequency** field for an exact coordinated MHz.

WATCH OUT

- ULX-D has no bodypack/portable receiver — all receivers are rack/half-rack.
- Re-sync after enabling/disabling **encryption**. Mixed bands are OK if both units can tune the same frequency (Shure “Mixed-Band Operation”).

SOURCES [ULX-D user guide](#)

Shure Axient Digital

Digital UHF · portable receiver shown

FREE-TUNE TO ANY MHz

Step 25 kHz

Range US: G57 470–616 (608–614 gap) · K54 606–663 · X55 941–960 — intl: G55/G56 470–636 · K57 606–790 MHz

Typical report label: **Shure AD G57** **Shure AD G57+** **Shure AD K54** **Shure AD X55** **Shure AD K57 (intl)**

TRANSMITTER

AD1/ADX1 · AD2/ADX2 (pack/handheld)

- 1 On the TX, **0** → **Radio** → **Freq**; **0** once edits the first 3 digits, twice the last 3; **AV** to set the assigned **MHz**; **0** to save.
- 2 Or IR-sync from the receiver (right). ADX can also be re-tuned remotely via ShowLink / AD610.

PORTABLE RECEIVER

ADX5D (2-ch portable receiver)

- 1 From the channel's home screen press **0** → **Radio**.
- 2 **AV** to set the assigned **Frequency (MHz)** directly (or Group/Channel); **0** to save.
- 3 Power the TX; channel menu → **Sync** (or hold **X** → **Rx1/Rx2 Sync**); align IR windows until the sync LED lights red → **Success**.

RF OFF FIRST

Hold **X** while powering on → **SAFE START** menu → set **RF: Mute** (persists through power cycles). Tune/sync, then un-mute: **Radio** → **Output** → **On**.

RF POWER

AD 2 / 10 / 35 mW · ADX 2 / 10 / 40 mW (Radio → Power; some bands 2 / 10 only; 10 mW max above 608 MHz).

MATCHING TX & RX

Channels are grouped for intermod safety, but you can type an exact **Frequency** on either device.

WATCH OUT

- Receiver **RF Band** must equal the TX band; after changing RF band, re-sync every TX.
- ADX5D is the portable receiver; AD4D/AD4Q are the rack versions.
- **G57+** (ADX5D) extends G57 with 614–616 MHz; TX power is capped at 10 mW there.

SOURCES [ADX1 guide \(Safe Start, tuning\)](#) · [ADX5D guide \(sync, bands\)](#) · [AD frequency chart](#)

Shure UHF-R

Analog UHF (legacy — still everywhere) · UR1/UR2 TX · UR4D rack RX

FREE-TUNE TO ANY MHz**Step** 25 kHz — strict grid, no fine offsets**Range** US: G1 470–530 · H4 518–578 · J5 578–638 · L3 638–698 — **614+ MHz is post-repack illegal in the US**Typical report label: **Shure UR4D H4** **Shure UHF-R J5****TRANSMITTER****UR1 / UR2 (bodypack / handheld)**

- 1 Press **enter** from the main screen → **Group (G) / Channel (Ch) or Frequency**; **▲▼** in 25 kHz steps; **enter** to save. (UR2: pull the battery cover off the handle to reach the keys.)
- 2 Or IR-sync: open the TX battery cover to expose the IR port; on the receiver select **Sync ▶ sync**.
- 3 A **frequency-locked** TX silently ignores the synced frequency — unlock first (hold **exit** + tap **▲** toggles Frequency Lock).

RECEIVER (RACK)**UR4D / UR4S (rack receiver — no portable in-series)**

- 1 **Radio** menu: push the **control wheel** to the Group/Channel or frequency value; turn to change; press the flashing **ENTER**.
- 2 Or **Radio ▶ Scan** → Chan Scan / Group Scan.
- 3 Then IR-sync down to the TX (step 2, left).

RF OFF FIRST**RF Safety Mode:** hold the **exit** key **while switching the TX on** → LCD flashes; it will not radiate while you set group/channel. Power-cycle to exit.**RF POWER**Hold **exit** + tap **▼** on the TX. UR2: **10 / 50 mW**; UR1: **10 / 100 mW** (G1/H4/J5; L3 = 10/50).**MATCHING TX & RX****Sync ▶ Setup** can also push gain, RF power and locks with the frequency — check what it's configured to send before syncing someone else's pack.**WATCH OUT**

- Analog FM on a strict 25 kHz grid — a .0125 offset doesn't exist here.
- Upper J5 and all of L3 (614–698 MHz) are illegal post-repack — check the band label before accepting the unit.
- Tonekey squelch (**Radio ▶ Squelch ▶ Tonekey**) should stay On.

SOURCES [UHF-R user guide](#) · [UHF-R spec sheet](#)

Shure PSM 1000 (IEM / IFB)

Analog FM (networked control) · the bodypack IS the portable receiver

FREE-TUNE TO ANY MHz**Step** 25 kHz**Range** Band on the label (e.g. G10 = 470–542 MHz)**Typical report label:** **Shure PSM-1000 G10****TRANSMITTER****P10T (rack transmitter)**

- 1 Press **ENTER** → select the channel (TX1/TX2) → **RADIO / Frequency** screen.
- 2 **ENTER** to edit; wheel/arrows to set the assigned **MHz** (or Group/Channel); **ENTER** to save.
- 3 Align the earpack IR window and press **SYNC** to push the freq to it.

PORTABLE RECEIVER**P10R+ (bodypack receiver, worn)**

- 1 Open the **battery door** — the menu buttons live inside. Press **ENTER** → **Group/Channel/Frequency**.
- 2 Edit the **Frequency** to the assigned **MHz**; confirm.
- 3 Or scan on the P10R+, then IR-**sync up** to the P10T.

RF OFF FIRST

Use the P10T's front-panel **RF switch** (per channel): RF off → power on → set the frequency → RF on ("Switch RF off and power on", per guide).

RF POWER

10 / 50 / 100 mW selectable (some bands top out at 50 mW).

MATCHING TX & RX

Here the **transmitter is the rack (P10T)** and the **receiver is the bodypack (P10R+)** — reverse of a mic.

WATCH OUT

- P10T and P10R+ must be the same band.
- Switch RF off before powering up near other systems.

SOURCES [PSM1000 user guide](#)

Sony UWP-D

Analog UHF + digital audio · portable receiver shown

PICK FINE GROUP FIRST**Step** 25 kHz raster**Range** Band on the label (UC14 470–542, UC25 536–608...)Typical report label: **Sony UWP (UTX-B40 / M40 / P40)****PICK FINE GROUP FIRST**

To reach **any** coordinated MHz, both units must be in a fine group: the group whose number **matches the TV channel** (e.g. Group **14** for TV-14) — these step in **25 kHz**. The low groups **01–09** are intermod presets with only a few fixed channels and will NOT reach an arbitrary freq.

TRANSMITTER**UTX-B40 / M40 / P40 (TX)**

- 1 Power OFF, then hold **SET** while powering on → **transmission-stopped** mode (RF off).
- 2 **+**/**−** to **GP/CH**; hold **SET** until **GROUP** flashes → pick the **TV-channel-number group** (not 01–09); **SET**.
- 3 **CHANNEL** flashes → **+**/**−** to the channel = assigned **MHz**; **SET**. Power-cycle to transmit.

PORTABLE RECEIVER**URX-P40 (portable diversity tuner)**

- 1 **+**/**−** to **GP/CH**; hold **SET** to edit. Select the same **TV-channel-number group**, then the channel matching the **MHz**.
- 2 Confirm the MHz on the OLED.
- 3 Or run **NFC / IR SYNC (RX→TX)** to copy the receiver's freq to the transmitter.

RF OFF FIRST

You're already RF-off: GP/CH is set in **transmission-stopped mode** (hold SET+POWER). Only power-cycle to transmit once the freq is right.

RF POWER

Menu **RF POWER: HIGH / LOW** — UTX-B40/M40 **30 / 5 mW**; UTX-P40 (US) **40 / 5 mW**; E/KR models 10 / 2 mW.

MATCHING TX & RX

Use Sony's **UWP-D Frequency List** to map an assigned MHz → Group/Channel. Group 00 = coarse 125 kHz steps. Tip: hold **NFC SYNC** ≥ 4 s = **Clear Channel Scan** of the selected group.

WATCH OUT

- Stuck on only a few channels = you're in a preset group (01–09); switch to the TV-channel group.
- GP/CH & BAND change only with transmission stopped.

SOURCES [UWP-D operating instructions](#) · [UWP frequency lists](#)

Sony DWX Digital

Digital UHF · portable (slot) receiver shown

FREE-TUNE TO ANY MHz**Step** 25 kHz (US 375 kHz channel plan)**Range** TX block must match the RX (e.g. TV14-17 = 470-494 MHz)**Typical report label:** **Sony DWR / DWT ("TV1417", "DWT 15-075"...)****PICK FINE GROUP FIRST**

Two paths: **FREQ INPUT** (RX1/RX2 menu) dials an exact MHz digit-by-digit (last two digits 00/25/50/75) and offers to push it to the paired TX — or **Band → Group → Channel**: bands are TV-channel *blocks* (TV14-25, TV26-38...). **Group 00 = free tuning** across the unit's whole band in **125 kHz** steps (Sony doesn't intermod-protect it — fine when coordination owns the plan). Single-TV-channel groups (e.g. Group 21) step **25 kHz** within one TV channel; D1-D7 / 01-07 are intermod presets.

TRANSMITTER**DWT-B03R / B01N / P-series (TX)**

- 1 Power OFF, hold **SET** and turn on → setting mode (transmission stopped).
- 2 **+**/**-** to **GP/CH**; hold **SET** until **GROUP** flashes → pick **GP 00** for free tuning (any freq on the 125 kHz raster); **SET**.
- 3 **CHANNEL** flashes → set to the channel = assigned **MHz**; **SET**. Power-cycle to transmit. (Assignment on a .025/.075 offset? GP 00 can't land there — use the TV-channel group instead.)

PORTABLE RECEIVER**DWR-S03D (2-ch slot receiver)**

- 1 Open the channel menu → Group/Channel; **SET** to edit. Pick the TV-channel group, then the channel = **MHz**.
- 2 Or **FREQ INPUT** in the same channel menu: enter the exact assigned MHz digit-by-digit; accept the prompt to push it to the paired TX.
- 3 Or press **MENU SEL** + **SET** to auto-scan, then **Cross Remote** pushes the freq to the TX.

RF OFF FIRSTYou're already RF-off: GP/CH is set in **setting mode** (hold SET while powering on = transmission stopped). Power-cycle to transmit after the freq is set.**RF POWER****2 / 10 / 25 mW** selectable (DWT-B03R).**MATCHING TX & RX**Cross Remote / pairing carries the exact freq from RX to TX (pair: TX hold **POWER** + **-**, or RF REMOTE ► PAIRING). Use the Sony **DWX Frequency List** for group mapping.**WATCH OUT**

- TX band/block must match the RX or they won't link.
- A small preset group shows only a few channels — move to the TV-channel group.

SOURCES [DWR-S03D manual](#) · [DWT-B03R manual](#) · [DWX frequency lists](#)

Wisyscom MTP / MCR

Wideband UHF · portable receiver shown

PICK FINE GROUP FIRST**Step** fine (to 25 kHz)**Range** Wideband, varies by model — MCR42-N 470–700 · MTP40S B7 470–663 · MTP60-US 470–608 / 614–663 / 940–960**Typical report label:** **Wisyscom wideband (e.g. 470–700)****PICK FINE GROUP FIRST**

Frequencies live in a **Group (GR 0–39)** / **Channel (CH 0–59)** grid; you edit the frequency inside the selected slot. Channels can be **locked** (lock icon = freq not editable; locking/hiding is managed in Wisyscom Manager) — pick an **unlocked** group/channel to free-tune.

TRANSMITTER**MTP40S / MTP60 (bodypack TX)**

- 1 MTP60:** **SEL/ON** + **GAIN/UP** opens the MENU → **RF ▶ Freq** (or short-press **CH** from the Status screen). Pick an **unlocked** GR, then CH.
- 2** At **Freq**, **SEL** to edit, **UP** / **DOWN** to set the **MHz**, **SEL** to save.
- 3 MTP40S:** use the scroll **selector** / **CH** keys → set **GR** (unlocked) → **CH** → **Freq**.

PORTABLE RECEIVER**MCR42 / MCR54 (portable receiver)**

- 1** Menu → **Edit RX: GR-Ch** (MCR54: Edit RX1–RX4) → select an **unlocked** group + channel.
- 2** Or **Edit RX: Frequency** to set the assigned **MHz** directly; confirm.
- 3** Or **IR-sync**: enable **IRDA on the TX first**, then hold the receiver's **sync (arrow-up)** button at the TX IR port.

RF OFF FIRST

Power up with transmission disabled — **MTP60:** hold EXIT/OFF to switch on without RF (or set RF Power = **OFF**); **MTP40S:** leave the wireless switch off. **LED2 RED = RF off**; set the freq, then enable a power level — **LED2 GREEN = radiating**.

RF POWER

MTP60: **10 / L10 / 20 / L20 / 50 / 100 mW** (+ OFF). MTP40S: **10 / 20 / 50 / 100 mW**. (RF ▶ Power)

MATCHING TX & RX

If the freq won't change, the active group/channel is **locked** — switch to an unlocked one (locks are managed in Wisyscom Manager).

WATCH OUT

- Enable RF — wireless power switch to **I** (LED green). Set the correct battery type.
- Keep one open group for field tuning; locks live in Wisyscom Manager.

SOURCES [MTP60 manual](#) · [MTP40S manual](#) · [MCR42 manual](#)

Lectrosonics Digital Hybrid

Block UHF · portable (slot) receiver shown

FREE-TUNE WITHIN BLOCK

Step 25 or 100 kHz — selectable on the unit (Channel Spacing setup)

Range Fixed BLOCK on the housing (470, 19, 21, 22...)

Typical report label: **Lectro 100 kHz Block 19** **Block 21** **Block 22** **Lectro 25 kHz Block 470**

FREE-TUNE WITHIN BLOCK

No menus full of presets — each unit free-tunes **within its hardware BLOCK**. The only “group” rule: **TX and RX must be the same block**. Confirm the block (engraved on the housing) before tuning.

TRANSMITTER

SMV / SMQV (and SMA, SSM, HMa plug-on...)

- 1 From home, press and **HOLD** **FREQ**; while holding, **UP**/**DOWN** moves the freq in 100 kHz steps. Release to set.
- 2 Press **FREQ** again to toggle the display between **MHz** and the 2-digit **hex** channel.
- 3 To set with RF off, boot Standby: with power OFF, briefly press **AUDIO** + **FREQ** together — boots showing “**rf OFF**”.

PORTABLE RECEIVER

SRc / SRc5P (2-ch slot receiver)

- 1 Press **MENU/SEL** to open setup; pick the receiver (1 or 2).
- 2 Hold **MENU/SEL** + **UP**/**DOWN** to step the **frequency** (or use **SCAN**, then set).
- 3 Then trigger **IR SYNC** to copy the receiver’s freq to the transmitter.

RF OFF FIRST

Boot in **STANDBY** (RF off): with power OFF, briefly press **AUDIO** + **FREQ** together — powers up showing “**rf OFF**”. Set the freq, then leave Standby to transmit.

RF POWER

50 / 100 / 250 mW in the Power menu (250 mW not on the 941 band).

MATCHING TX & RX

Block = the fixed hardware range; it can’t be changed in the field on single-block units. Match TX and RX blocks. (The SRc is itself a 3-block wideband RX — set its **STEPSIZE** 25/100 to match.)

WATCH OUT

- If TX and RX blocks differ, they will never link.
- Spacing defaults to 100 kHz — set **Channel Spacing = 25 kHz** (setup menu / LectroRM) to reach .025 / .050 / .075 assignments. Hex is just a display format.
- Blocks run past TV-band UHF — Block 944 (legacy, 944.100–951.900) and current **-941** variants (941.525–951.975) = the STL band; same tuning drill.

SOURCES [SMV/SMQV manual \(Standby\)](#) · [SRc manual](#)

Lectrosonics L-Series wideband

Digital Hybrid UHF · spans 3 legacy blocks · portable receiver shown

FREE-TUNE WITHIN BLOCK

Step 100 kHz default — set Step Size = 25 kHz

Range A1 470.100-537.575 · B1 537.600-607.950 (*US TX; RX tunes to 614.3*) · C1 614.400-691.175 (each unit covers one 3-block range)

Typical report label: **Lectrosonics 3 Block A1** **Lectrosonics 3 Block B1** **LT / LMb → LR**

FREE-TUNE WITHIN BLOCK

Same Digital Hybrid family as the block-units card, but each unit free-tunes across **three legacy blocks** (“3 Block” on reports). The catch is the **Step Size** setting: front panels default to **100 kHz** — set **Step Size = 25 kHz on BOTH ends** or a .025 / .050 / .075 assignment is unreachable.

TRANSMITTER

LT / LMb / SMWB / SMDWB (wideband bodypacks)

- 1 Power on in **Standby Mode**: short-press the power button, releasing *before the countdown reaches 3* — RF stays off (“Standby” in the Main Window) while you tune.
- 2 Menu → **StepSize**: set **25 kHz**. Then **Frequency**: **UP**/**DOWN** to the assigned **MHz**.
- 3 **Rf On** from the Main / Power menu once confirmed — or IR-sync from the LR first, then enable RF.

PORTABLE RECEIVER

LR (camera-mount receiver)

- 1 **MENU/SEL** → set **Step Size = 25 kHz**, then open **Frequency** (or run **SmartTune** to scan for clear air).
- 2 Dial the same assigned **MHz**; confirm on the LCD.
- 3 Trigger **IR sync** at the transmitter (RX → TX) — it carries **frequency, step size and compat mode** in one go.

RF OFF FIRST

Power the LT / LMb on in **Standby Mode** (RF off) and do all tuning there; turn **Rf On** from the Main / Power menu once set. **AutoOn** in the Power menu controls whether RF comes up at boot.

RF POWER

50 / 100 mW (Power menu).

MATCHING TX & RX

TX and RX ranges must overlap on the assigned MHz (A1 / B1 / C1 spans). **Step Size must match on both ends** — same rule as the block-units card, wider hardware.

WATCH OUT

- Reports list these as **“Lectrosonics 3 Block A1 / B1”** — the wideband LT / LR / SMWB family (SMWB/SMDWB tune the same way; “Nu Hybrid” on a report = these in Digital Hybrid compat), not the single-block SMV card.
- Step Size left at 100 kHz silently hides .025 / .050 / .075 assignments — set **25 kHz** on both TX and RX.
- Compat mode (Digital Hybrid / 100 / 200 / IFB / Mode 3...) must match the receiver or no audio.

SOURCES [LT manual](#) · [LR manual](#)

Lectrosonics IFB (IFBlue)

Block UHF/VHF IFB — talent & crew cue feeds · portable receiver shown

FREE-TUNE WITHIN BLOCK**Step 25 / 100 kHz (VHF 175 kHz)**

Range Fixed ~75 MHz range; RX & TX ranges must match (A1 470–537, B1 537–614, C1 614–692; VHF 174–216; “-941” 941.5–952)

Typical report label: **Lectro IFBT4** **IFBT2.0** **Digital Hybrid TX in IFB mode**

FREE-TUNE WITHIN BLOCK

IFB is one-way: an IFB transmitter feeds belt-pack receivers. No presets to fight — each unit free-tunes within its **range/block**. Rule: the **receiver's range must contain the transmitter's frequency**, and both must speak the same **compatibility mode** — set on the TX (IFBT4: IFB default / 100 / 200 / Mode 3/6; IFBT2.0 adds NU/EU Hybrid). The IFBlue RX has no compat menu.

TRANSMITTER

IFBT4 / IFBT2.0 (or any Lectro DH TX in IFB mode)

- 1 IFBT4:** set the switch to **TUNE** (RF off — freq changes only in TUNE). Set **COMPAT** to match the RX (IFB is the default).
- Press **MENU** to the **Frequency** window; **UP**/**DOWN** to the assigned **MHz** (**MENU**+**UP**/**DOWN** jumps 16 ch). TUNING = **NORMAL** for free choice.
- Flip to **XMIT** to go on air. (**IFBT2.0:** menu-driven colour LCD — pick the channel's Frequency + mode within its A1/B1/C1 range.)

PORTABLE RECEIVER

IFBlue IFBR1C / IFBR1B (belt-pack)

- 1** Press **FREQ** to show the receiver frequency in **MHz**.
- 2** **UP**/**DOWN** to the assigned **MHz** (25 or 100 kHz steps; VHF 175 kHz). **FREQ**+**UP**/**DOWN** jumps 1 MHz.
- 3** Store it to one of the **10 presets**; **UP**/**DOWN** then scrolls saved freqs on the fly. (IFBR1C tunes the same way in a tinier body.)

RF OFF FIRST

Set the IFBT4 switch to **TUNE** — RF is off (a blinking “T” shows) and the freq can only be changed here. Flip to **XMIT** once set.

RF POWER

IFBT4: ~250 mW. IFBT2.0: selectable **10 / 25 / 50 / 100 / 250 mW**.

MATCHING TX & RX

No sync. Set the **same MHz** on TX and RX, keep both in the same **compatibility mode**, and make sure the RX range covers the TX frequency. A **pilot tone** keeps the RX muted until it sees a valid signal.

WATCH OUT

- RX and TX must share the tuning range (A1/B1/C1 or VHF) or it can't tune there.
- Hearing nothing? Check COMPAT mode and the pilot tone — not just the frequency.
- IFBT4 frequency changes only in TUNE, not XMIT.

SOURCES [IFBT4 manual](#) · [IFBR1B manual](#) · [IFBT2.0 product page](#)

Lectrosonics Duet (M2T / M2Ra)

Digital UHF IEM / IFB · the bodypack IS the portable receiver

FREE-TUNE TO ANY MHz

Step 25 kHz

Range A1B1 470.100-607.975 (NA) · B1C1 version 537-691 — the unit's range must cover the assigned MHz

Typical report label: **Lectrosonics M2T A1B1**

TRANSMITTER

M2T (half-rack TX — two carriers, two audio ch each)

- 1 **Menu** → **RF Tuning** → pick the carrier (each of the two carriers tunes separately); **UP**/**DOWN** in **25 kHz** steps; confirm.
- 2 **RF Enable/Level**: keep the carrier **off** while tuning; set the power level per carrier.
- 3 Or drive the whole rack from **Wireless Designer** (USB / Ethernet) — freq, names, levels.

PORTABLE RECEIVER

M2Ra (belt-pack stereo receiver)

- 1 Open the menu and tune to the assigned **MHz** (25 kHz steps) — or run **SmartTune** to scan on-site RF.
- 2 **2-way IR sync**: hold the M2Ra to the M2T's IR port — settings and scan data transfer between units (RX → TX works).
- 3 If the rig uses a **FlexList**, pick the feed by name instead of dialing.

RF OFF FIRST

The **RF Enable/Level** menu keeps each carrier OFF until you explicitly enable it — tune first, enable second. Same habit as the SMV standby boot, just menu-driven.

RF POWER

10 / 25 / 50 mW per carrier (RF Enable/Level menu).

MATCHING TX & RX

Transmitter is the rack (M2T), receiver is the belt-pack (M2Ra) — reverse of a mic, like PSM 1000. 2-way IR sync moves settings in either direction.

WATCH OUT

- Not the IFB card: Duet is the **digital stereo** IEM system. The M2Ra's **IFB / D2 / HDM** compat modes let it stand in for an IFB R1a or D-Squared receiver — set the mode to match the TX.
- Two independent carriers per M2T half-rack — confirm *which* carrier you're tuning before you dial.
- A1B1 vs B1C1 hardware versions — the assigned MHz must sit inside the unit's range.
- M2Ra **IFB compat mode is mono only**.

SOURCES [M2T manual](#) · [M2Ra manual](#)

Lectrosonics D Squared (DBSM / DSQD)

Pure digital · DBSM/DBSMD pack TX · DSQD rack / DSR4 slot RX

FREE-TUNE WITHIN BLOCK

Step 25 kHz — always; no Step Size setting to trip on

Range US DBSM(D)-A1B1 **470.100-607.950** · DSQD/DSR4 tune 470.1-614.375 (NA locale blocks 608-614)

Typical report label: **Lectro DBSM A1B1** **Lectro DSQD / DSR4**

FREE-TUNE WITHIN BLOCK

Same block rule as the rest of the Lectro family: the TX's **range variant** (A1B1 / B1C1) must contain the assigned MHz. Unlike Digital Hybrid there is **no 100 kHz default** — everything steps 25 kHz.

TRANSMITTER**DBSM / DBSMD (membrane keypad)**

- 1 Short-press** the power button = **Standby** (RF off, indicator blinks) — do all tuning here.
- 2 TopMenu ▶ Xmit ▶ Freq;** **MENU/SEL** toggles the MHz / kHz fields; **▲▼** steps 1 MHz / 25 kHz.
- 3** Enable RF with a long press **through the 3-count** (or **Xmit ▶ Rf On? Yes**).

PORTABLE RECEIVER**DSQD (rack) / DSR4 (4-ch slot receiver)**

- 1 RF Setup ▶ RF Frequency** (**MENU/SEL** + **▲▼**); or **Smart Tune** / Frequency Scan.
- 2 IR Sync & Encryption ▶ Sync Settings → SEND FREQ** (or SEND ALL) → pick channel 1-4 → **GO**.
- 3** Hold the DBSM's IR port to the receiver — the TX has **no sync menu**; it just listens → "SUCCESS".

RF OFF FIRST

Built into the power button: **short press = Standby (RF off)**; releasing a long press *before the countdown reaches 3* also boots RF-off. Full 3-count (or Xmit ▶ Rf On?) to radiate.

RF POWER

Xmit ▶ TxPower: 10 / 25 / 50 mW, or **HDM** = 2 mW high-density mode.

MATCHING TX & RX

Compat modes must match end-to-end (D2 vs HDM; DSQD/DSR4 also decode Duet CH1/CH2 and Hybrid modes). Wireless Designer drives the receiver; the TX is reached only via IR.

WATCH OUT

- Encryption **key-type** mismatch = RF meters but no audio.
- Never set DSQD/DSR4 channels 1+3 or 2+4 to the same frequency.
- HDM = 2 mW only** — range-check the walk before committing.
- Exit the DSQD setup screen before powering down or settings aren't saved.
- DCHT** camera-hop TX is family: wideband 470.1-607.950, 25 kHz, **Xmit ▶ Freq**; pairs with DCHR (native **DCHX** mode) or DSQD (fw ≥ 1.04); 10 / 25 / 50 mW.

SOURCES [DBSM/DBSMD manual](#) · [DSQD manual](#) · [DSR4 manual](#) · [DCHT manual](#)

Comtek 216 (BST-25 / PR-216)

VHF FM IFB / cueing (216–217 MHz) · belt-pack receiver shown

CHANNEL CHART ONLY

Step channel chart — 57 channels

Range 216–217 MHz (FCC Parts 22/74/90) — 38 narrow + 19 hi-fi companded channels

Typical report label: **Comtek BST-216 Narrow 23****CHANNEL CHART ONLY**

No MHz dial at all — coordination assigns a **channel from Comtek's published chart** (e.g. "Narrow 23" = 216.5625). Narrow (non-companded) and hi-fi (companded) are **separate channel sets**; the TX auto-sets deviation / EQ per channel, and the RX must be on the same channel type.

TRANSMITTER**BST-25/216 (base-station transmitter)**

- 1 Select the assigned **channel** with the front-panel controls — the LED window shows the channel *and* exact frequency.
- 2 Confirm the channel **type** (narrow vs hi-fi companded) matches the assignment; deviation/companding follows the channel — but **Speech EQ is a manual rear-panel switch**.
- 3 Switch the **DISPLAY off** once set — with the display disabled the tuning controls are locked, so the freq can't walk.

PORTABLE RECEIVER**PR-216 (belt-pack receiver)**

- 1 Select the **same channel number** on the PR-216 via the two rear rotary switches (tens/ones) — it tunes both channel sets.
- 2 Use the BST's **400 Hz TONE** switch to confirm the link and set listen level, then kill the tone.
- 3 Channel stays put once set — no sync, nothing else to match.

RF OFF FIRST

No RF-off standby — the BST radiates once powered. **Set the channel before powering near the rig** (or before the antenna goes on), then use the display-off lockout to freeze tuning.

RF POWER

BST-25: fixed, **set for 100 mW** (FCC Parts 22/74/90). BST 75-216 = compact 10-channel "Mini Base Station" with *selectable* power; M-216 is the belt-pack TX.

MATCHING TX & RX

No sync — match the **channel number** on TX and RX from the chart. Running 2+ channels on one site needs intermod-checked channel groups (Comtek's chart / frequency software).

WATCH OUT

- VHF 216–217 — completely outside UHF TV coordination; it appears on reports as a **channel number**, not a free MHz.
- Narrow vs hi-fi companded are different channel sets — a mismatch passes RF but sounds wrong.
- Display disabled = tuning locked (the rear **LOCK OUT** switch does the same). Re-enable to change channels.

SOURCES [BST 25-216 manual](#) · [PR-216 manual](#)

Sound Devices Astral (A20)

Digital wideband (SpectraBand) · portable receiver shown

FREE-TUNE TO ANY MHz**Step** 25 kHz (off the sub-channel grid is OK)**Range** SpectraBand 470–1525 MHz (A20-TX & A20-RX reach down to 169; HH/Mini start at 470) — region/license-gated

Typical report label:

Astral TX (A20-TX)**Astral Mini (A20-Mini)****Astral HH (A20-HH)****TRANSMITTER****Astral TX / HH (display) · Mini (app)**

- 1 Astral TX / HH:** open the menu → **Frequency**. (There is no Country menu on the unit — region limits are set automatically via the **A20-Remote** app or an Astral receiver.)
- 2** Dial the assigned **MHz** and confirm; turn **RF on**.
- 3 Astral Mini** (no screen): set frequency from the **A20-Remote app**, or push it from the receiver / Nexus over **NexLink**.

PORTABLE RECEIVER**A20-RX / ARX2 (2-ch portable receiver)**

- 1** Press  or  to select a channel (single-channel view); press the centre **ENTER** to open its menu.
- 2** Go to **Frequency** → **ENTER**;   to set the assigned **MHz**; **ENTER** to confirm. (An * just means it's off the preset sub-channel grid — fine.)
- 3** Set the transmitter to the **same MHz** (on-unit, app, or NexLink). Tip: **AutoAssign** scans a chosen band and fills both channels.

RF OFF FIRSTUse the menu **RF Off** toggle (or set RF Power = **Off** on the Mini) so it's powered but not radiating; set the freq; then **RF On**.**RF POWER**A20-TX/Mini **2 / 10 / 20 / 40 mW** (40 needs the **Extra RF Power** enable); HH adds 40 “Very High” / **100 mW “Extra High”**, both behind the same enable. Set on the unit, A20-Remote, or the receiver. Tip: raise Long-Range modulation before power.**MATCHING TX & RX****No IR sync.** TX and RX must share the same **frequency + modulation**; match by hand, via **A20-Remote**, or push over **NexLink** (the screenless Mini must be set this way).**WATCH OUT**

- Available freqs are country-gated (set via A20-Remote / receiver, plus any **FA license code**); restricted bands stay locked.
- GainForward:** no gain on the transmitter — gain is set at the receiver / mixer.
- Digital & intermod-immune, but keep A20 units ≥ **400 kHz** apart.

SOURCES [A20-TX user guide](#) · [A20-RX user guide](#) · [Frequencies by country](#)

Zaxcom (ZMT4.5 / URX100)

Pure digital · ZaxNet 2.4 GHz remote control · no IR anywhere

FREE-TUNE TO ANY MHz**Step** 100 kHz (0.1 MHz per INC/DEC press)**Range** ZMT4.5 **512-608** (Blocks 20-23) · ZMT4.6 578-698 · VTX1 VHF 192-217 (FM 215-216.975 = Comtek-compatible)

Typical report label:

Zaxcom ZMT4.5 (ZHD48)**Zaxcom VTX VHF****Zaxcom TRXCL5****TRANSMITTER****ZMT4.5 (micro bodypack TX)**

- 1 From Home press **MENU** to **“UHF Transmit Frequency Set”** (TXFREQ).
- 2 **INC**/**DEC**: short press = 0.1 MHz, hold = 0.5 MHz jumps. Direct dial — no groups, no banks.
- 3 Keys locked? Hold **MENU** + press **INC** ×5 to unlock.

PORTABLE RECEIVER**URX100 (portable confidence RX; bags run Nova/MRX214)**

- 1 **MENU** to **“Frequency Select”**; **CH UP**/**CH DOWN** tune in 0.1 MHz.
- 2 Or **Frequency Scan** → accept the suggested clear freq with **CH UP**.
- 3 **Modulation must match the TX** — URX100 decodes ZHD96 / XR / MONO but **not ZHD48** (ZHD48 needs a current MRX214).

RF OFF FIRST

Setup ▶ “Standby Mode After Boot Up”: **STANDBY** (wakes on MENU press) or **REMOTE STANDBY** (boots RF-silent, waits for a ZaxNet **WAKE** from Nova2 / VTX1). **Transmit ▶ Record-Only Mode** kills UHF entirely.

RF POWERZMT4.5 **TX POWER: 10 / 50 mW**. VTX1: 10 / 25 / 50 mW.**MATCHING TX & RX**

No IR. Frequencies can be pushed over **ZaxNet 2.4 GHz** (Group Code 1-99 + Unit Code 1-200 address each pack; controller = Nova2 “TRX Frequency” or VTX1 remote menu).

WATCH OUT

- Modulation mode (ZHD48 / ZHD96 / XR) must match TX↔RX — changing it forces a reboot.
- ZHD48 occupies ~50 kHz and can sit 100 kHz apart — expect tighter packing than analog on the report.
- ZaxNet (2.403-2.475 GHz) is a real on-site emitter — coordinate it too.
- **TRXCL5** camera link tunes the same way (INC/DEC, 100 kHz): US 512-607.9 up to 100 mW (+ 614.1-615.9 / 653.1-662.9 at 20 mW); power 25 / 50 / 100 mW.

SOURCES [ZMT4/4.5 manual](#) · [URX100 manual](#) · [VTX1 manual](#) · [TRXCL5 manual](#)

Shure Wireless Workbench (WWB 6 & 7)

Software · making WWB accept coordinator-assigned frequencies · nothing on this card transmits

SOFTWARE — CALC ONLY

Step profiles: Robust · Standard · More Frequencies

Range applies only to the WWB calculator on your laptop — it changes nothing on any transmitter

Typical report label: **"WWB rejects our assigned freqs"** **Compatibility profile**

SOFTWARE — CALC ONLY

Coordination hands you exact frequencies; WWB flags them (intermod / "3TO" warnings). **WWB isn't overruling the event plan — it's re-running its own intermod math with conservative margins** that the site-wide coordination already handled. Relax the **compatibility profile** so WWB accepts what you were given — do **not** calculate new frequencies.

WWB 6 WWB 6

- 1 **Frequency Coordination** tab → click the device-type **group header** (its text shows the active profile).
- 2 Right sidebar → **Compatibility** tab → **Compatibility Profile** drop-down → **More Frequencies** (try **Standard** first if you only need a little slack).
- 3 Keep the assigned freqs; **Calculate** — the warnings should clear. (Spacing numbers live in **Tools ▶ Equipment Profiles**.)

WWB 7 WWB 7

- 1 **Frequency coordination** workspace → click the device **header** (dark grey bar); **Ctrl**-click to select several.
- 2 **Compatibility** tab → profile drop-down → **More frequencies** (or **Standard**). Fine sliders: **Channel / Intermod spacing** (2- & 3-TX); **Revert** undoes, **Save** keeps a modified profile (*).
- 3 **Calculate**. Global default: **Preferences ▶ Coordination ▶ Compatibility** → double-click the model's **Coordination Preference**.

DON'T RECALC

Type the coordinator's frequencies **exactly** (double-click the Inventory Frequency column). Relaxing the profile only makes WWB *accept* them — **don't generate new frequencies** without asking coordination.

TRADE-OFF

More Frequencies = tighter spacings: less margin against intermod products and short TX-to-RX distances. Fine when the site-wide coordination owns the plan; **not** a free win when you're self-coordinating in isolation.

WATCH OUT

- "3TO"-style warnings are WWB's own 2- and 3-transmitter third-order intermod spacing rules — a margin policy, not a verdict on the event plan.
- Set the profile **per device type** — relax only the types you were handed frequencies for.
- If WWB still refuses one specific assigned frequency, don't nudge it to a nearby number — **call coordination**; the digits are load-bearing.

SOURCES [WWB help — Adjust Compatibility Settings](#) · [WWB6 compatibility profiles \(Shure\)](#) · [WWB help — Specify a Frequency](#)

Teradek Bolt / Ranger (wireless video)

5 / 6 GHz zero-delay video · lock it to one channel so it can't hop

LOCK TO ONE CHANNEL

Step 20 MHz per link (factory default is 40 MHz — twice the footprint)

Range 5 GHz U-NII-1/2/3 · 6 GHz U-NII-5 5.925–6.425 (Bolt 6 · Bolt 4K LT-HDMI · Ranger Mk II/Micro)

Typical report label:

Teradek Bolt 6 (6 GHz)

Teradek Bolt 4K (5 GHz)

Teradek Ranger

LOCK TO ONE CHANNEL

Out of the box a Bolt **auto-hops across the whole band at 40 MHz**. On a coordinated site it must sit still: **Fixed Frequency on, one assigned channel, 20 MHz** — set identically on **both TX and RX** (“ensure that both the transmitter and receiver have Fixed Frequency Mode enabled”).

LOCK THE FREQUENCY

Same steps on TX and RX

- 1 **Menu** → **Wireless Settings** → **Enable Fixed Frequency** — on **both** TX and RX (“bypasses any automatic frequency switching logic”).
- 2 Then **Wireless Settings** ▶ **Frequencies**: select **only the assigned channel**. If you skip this, Bolt silently defaults to the **lowest** frequency in the list.
- 3 Fixed mode offers **non-DFS channels only** — if your assignment is a DFS channel, the unit cannot lock there; call coordination.

LOCK 20 MHz + BAND

Same steps on TX and RX

- 1 **Menu** → **Wireless Settings** → **Bandwidth** → **20MHz** (default 40) — both ends must match. 20 MHz carries up to **1080p60**; 4K59/60 needs 40 MHz.
- 2 6 GHz-capable units: **Wireless Settings** ▶ **Band** → 5GHz or 6GHz per the assignment — both ends; hopping never spans bands.
- 3 Sanity-check with the **Spectrum Analyzer** (RX front panel / Bolt app). Changed the Region setting? **Re-pair** TX and RX.

NO DFS LOCK

Fixed Frequency = non-DFS only. DFS ranges (5.25–5.35 / 5.47–5.725 GHz) need a 1-minute radar scan and can be forced to vacate — don't assign video links there.

FOOTPRINT

6 GHz (U-NII-5) runs at **lower allowed power** than 5 GHz but has **no DFS**. At 20 MHz a Bolt 6 offers **36 channel choices** vs 24 at 40 MHz.

MATCHING TX & RX

Ranger is the special case: spans 4.9–6.0 GHz including **licensed channels** — enable **Allow Licensed Channels** + Fixed Frequency to pick one (licensed operations only).

WATCH OUT

- Serv / Prism / Cube are **IP encoders on ordinary Wi-Fi** — nothing to lock on the unit; coordinate the network instead.
- Frequency, bandwidth, and band must match on **both ends** or the link won't form.
- See the companion **5/6 GHz channel chart** card for channel ⇄ MHz mapping.

SOURCES [Bolt 6 wireless configuration](#) · [Bolt 4K wireless configuration](#) · [Bolt 6 GHz support by region](#) · [Frequencies by region chart](#)

5 / 6 GHz Channel Chart (video & Wi-Fi)

Generic 20 MHz channelization · channel $\rightleftharpoons$ center MHz · who can go to 6 GHz

REFERENCE CHART

Center MHz = 5000 + 5×ch (5 GHz) · 5950 + 5×ch (6 GHz)

40 MHz = two adjacent 20 MHz slots

Use with:

Teradek Bolt / Ranger card

Venue Wi-Fi / IP video coordination

5 GHz BAND	SPAN (MHZ)	20 MHz CHANNELS (CENTER MHZ)	RULES
U-NII-1	5150–5250	36 (5180) · 40 (5200) · 44 (5220) · 48 (5240)	No DFS — lockable
U-NII-2A	5250–5350	52 (5260) · 56 (5280) · 60 (5300) · 64 (5320)	DFS — radar scan, can be evicted
U-NII-2C	5470–5725	100 (5500) · 104 · 108 · 112 · 116 · 120 · 124 · 128 · 132 · 136 · 140 · 144 (5720)	DFS — radar scan, can be evicted
U-NII-3	5725–5850	149 (5745) · 153 (5765) · 157 (5785) · 161 (5805) · 165 (5825)	No DFS — lockable , but shared with venue Wi-Fi
6 GHz BAND	SPAN (MHZ)	20 MHz CHANNELS (CENTER MHZ)	RULES
U-NII-5	5925–6425	1 (5955) · 5 (5975) · 9 (5995) ... step +4 ch / +20 MHz ... 93 (6415)	No DFS ; lower power (LPI); only band Teradek uses
U-NII-6	6425–6525	97 (6435) – 113 (6515)	Wi-Fi 6E/7 only — no Teradek products here; venue APs may be
U-NII-7	6525–6875	117 (6535) – 185 (6875)	
U-NII-8	6875–7125	189 (6895) – 233 (7115)	

6 GHz CAPABLE

Teradek units that can take a 6 GHz assignment: Bolt 6 XT (750 / 1500 / MAX / 10K RX) · Bolt 6 LT (750 / 1500 / MAX) · Bolt 6 LT-HDMI · Bolt 6 Monitor Module · **Bolt 4K LT-HDMI** (firmware $\geq$ 4.3.0 — the only Bolt 4K that can) · **Ranger Mk II** · **Ranger Micro**. Everything else Teradek is 5 GHz only.

WATCH OUT

- Teradek fixed-frequency locks only exist in the **green rows** (U-NII-1, U-NII-3, U-NII-5).
- A 40 MHz link eats **two** adjacent 20 MHz slots — count both when charting.
- Venue Wi-Fi lives in these same bands; 6 GHz (U-NII-5) is usually the cleanest room on site.

SOURCES [Teradek frequencies by region](#) · [Bolt 6 6 GHz support](#) · [WLAN channel list \(IEEE numbering\)](#)

2.4 GHz Auto-Hop Mics (DJI / Rode / Hollyland)

Consumer digital mics · adaptive frequency hopping · there is no frequency to dial

CANNOT BE COORDINATED

Step none — the radio picks its own hops, continuously

Range 2.400–2.4835 GHz ISM (DJI Mic 3 can also use 5.1–5.8 GHz)

Typical report label: **DJI Mic / Mic 2 / Mic Mini** **Rode Wireless GO II / PRO / ME** **Hollyland Lark M2 / Max**

CANNOT BE COORDINATED

These mics **hop across the whole 2.4 GHz ISM band automatically** (DJI: GFSK; Rode: “Series IV” digital; Hollyland: adaptive frequency hopping). **No menu, sync, or app setting pins them to a frequency** — on a coordination report they can only appear as a wideband 2.4 GHz occupant, never as an assigned channel.

WHAT THEY ARE

All 2.4 GHz ISM, all auto-hop

- 1 **DJI Mic / Mic 2 / Mic Mini:** 2.400–2.4835 GHz, GFSK, < 20 dBm EIRP (Mic 3 adds 5.1–5.8 GHz).
- 2 **Rode Wireless GO II / PRO / ME:** “Series IV” 2.4 GHz digital, 128-bit encryption, auto-pairing.
- 3 **Hollyland Lark M2 / Max:** 2.4 GHz adaptive frequency hopping, < 20 dBm EIRP.

WHAT TO TELL THE CREW

Kindly, at the check-in table

- 1 These can’t take an assigned frequency — **that’s the design, not a setting you’re missing.**
- 2 They share 2.4 GHz with venue Wi-Fi, Bluetooth, ZaxNet, and comms — count them on the walk anyway; every one raises the noise floor.
- 3 Power off units that aren’t actively shooting; if reliability matters, hand them a coordinated UHF pack instead.

SATURATION

Saturation is the endgame: each link needs clean hop-space. Cluster enough of them (plus Wi-Fi, Bluetooth, ZaxNet) in one small area and the band fills — **all of them start dropping at once**, and no settings menu can fix physics.

RF POWER

Fixed by regulation, not by menu: ISM-band devices run < 20 dBm EIRP and must tolerate each other’s interference (FCC Part 15). Range claims (100–400 m LOS) assume a quiet band — an event floor is not one.

HISTORY BONUS

Why they hop: frequency-hopping spread spectrum was co-invented in 1942 by actress **Hedy Lamarr** and composer George Antheil — [US Patent 2,292,387, “Secret Communication System”](#). Every mic on this card is her great-grandchild.

WATCH OUT

- Nothing on this card is field-tunable — if a crew insists their unit “has a channel setting,” it’s a pairing menu, not a frequency menu.
- DJI Mic 3 in 5 GHz mode lands in the same U-NII bands as Teradek video — see the 5/6 GHz chart card.
- These are fine tools in isolation; the problem is *density*.

SOURCES [DJI Mic 2 specs](#) · [Rode Wireless GO II](#) · [Hollyland Lark Max specs](#) · [Lamarr FHSS patent \(1942\)](#) · [Hedy Lamarr](#) — MIT Lemelson